Name: _____

WORKSHEET

# Representing vectors

**1** Use a ruler and protractor to represent the following displacements on the same set of axes.

**a** 1 cm east

**b** 4 cm south

**c** 3 cm west

**d** 2 cm north

**e** 5 cm north-west

**f** 5 cm south-east

**g** 3.5 cm north-east

**h** 2.5 cm south-west

**2** Use a ruler and protractor to represent the following displacements on the same set of axes.

**a** 3 cm east

**b** 7 cm NW

**c** 1 cm SE

**d** 5.5 cm SSW

**e** 3.5 cm ENE

**f** 4 cm WNW

**g** 5 cm NNE

**h** 4.5 cm SSE

**i** 2.5 cm WSW

**j** 4 cm NE

**k** 4.5 cm ESE

**l** 3.5 cm SW

**m** 5 cm NNW

**n** 3.5 cm west

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**3** Draw the following distances and bearings as position vectors on the same set of axes.

**a** 8 km N0°W

**b** 10 km N10°E

**c** 6 km N90°W

**d** 4 km N65°W

**e** 7 km S75°W

**f** 5.6 km S53°E

**g** 9.4 km N81°E

**h** 10.4 km S7°W

**i** 6.8 km S15°E

**j** 4.6 km S38°W

**k** 8.2 km N35°E

**l** 7.2 km S87°E

**m** 3.8 km N17°W

**n** 2.8 km S59°W

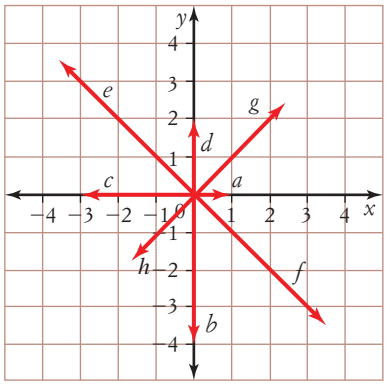
**o** 9.4 km N80°W

**p** 9.2 km N61°E

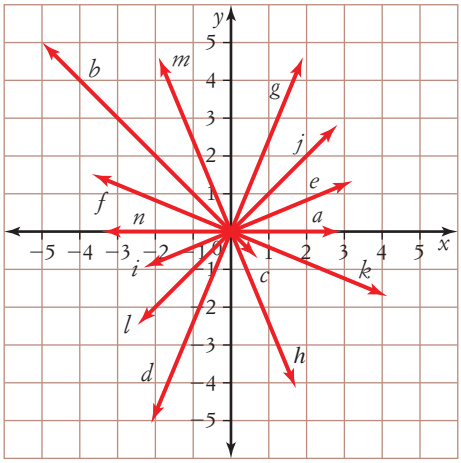
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## Answers

**1** 1 unit : 1 cm




**2** 1 unit : 1 cm




**3** 1 unit : 2 km

